

I accuse! Part one

The Computer Science Games (CSGames) 2024 CO worked hard to create the challenges, schedules, logistics, etc. After their efforts, they decided to relax and reward themselves by playing a classic game, CLUE.

Unfortunately, Colonel de Lachevrotière forgot the game box at home. Determined not to play another board game, Mrs Prevot came up with an idea: she proposes a new game that recreates the spirit of the CLUE game.

Game Setup

It's a one-on-one game, but an additional person (the dealer) is needed to set up the initial state of the game. Other members of the CO can only be spectators.

The game is played with a small envelope and a reduced deck of cards (specifically, only the hearts from a standard deck, so 13 cards).

The dealer performs the following operations out of sight of the two players:

- He randomly shuffles the deck.
- He randomly selects two cards from the deck and places them in the envelope. These cards are called the guilty cards.
- From the remaining cards, he randomly selects 5 cards and gives them to player 1.
- He gives the remaining 6 cards to player 2.

Of course, as in the real Clue game, all cards have been face down during these operations, so neither player knows what is in the envelope or in the other player's hand.

Game Rules

Players take turns playing. Player 1 goes first. On their turn, a player chooses ONE of the following two actions:

- Draw: **The player randomly draws a card from their opponent's hand.** Of course, the opponent can shuffle their cards behind their back before drawing to ensure that the drawing player is indeed drawing randomly. **Then, the player adds the drawn card to their hand.**
- Accusation: **The player says "I accuse" by naming the two cards they believe are in the envelope.** At this point, the game ends. The envelope is opened. The player wins if their statement is correct; otherwise, the opponent wins.

There is no limit to the number of rounds the game can last. Indeed, if you have understood correctly, a player always draws from a hand containing 6 cards. They can be really unlucky and often draw a card that was just taken from them (which provides no new information).

Your Mission

Calculate the probability that player 1 wins the game, assuming both players have an excellent memory and play optimally (i.e., only make an accusation if it is the best choice).

Flag

The file 'flag-steps-fake.txt' serves as an example to guide you on how to create the FLAG. Except for one big detail, reproduce exactly the steps in 'flag-steps-fake.txt'.

The big detail is, of course, that you must use the correct probability to perform these steps, instead of the $\frac{175}{375}$ provided as an example.

I accuse! Part 2

It seems that the previous version of the game does not satisfy player 1. Due to the calculation you made earlier, he complains that his chances of winning are not close enough to 50% for his liking.

To get closer, he proposes the following parameter changes:

- Play with all hearts AND all spades, so 26 cards, twice as many as before!
- The dealer puts 3 random cards in the envelope instead of 2. Thus, a player who accuses must guess the 3 cards.
- Players receive 11 and 12 cards respectively instead of 5 and 6.
- All other rules remain unchanged.

However, with these new parameters, other CO members are afraid of falling asleep. They claim that they will fall asleep if both players play more than 25 turns each.

Your Mission

Similar to the previous question, players still play optimally. No accusing just to please the spectators who will fall asleep!

Calculate the probability that the number of draws reaches at least 50. A draw is the act of randomly taking a card from the opponent's hand. For example, if both players take 25 turns each without accusing, that makes 50 draws.

Flag

Same format as the previous flag.